

assignment Newton_applications advanced -

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:36pm

Quiz Instructions

There are tutorials. Important for civil engineers/Mechanical/Physics.

Other majors. Some of them are challenging. Don;t worry too much.



Question 1 5 pts

Three vectors, $\vec{s}$, $\vec{T}$, and $\vec{U}$, have the components shown in the table. What is the magnitude of the resultant of these three vectors?

	x component	y component
$\vec{s}$	3.50 m	-4.50 m
$\vec{T}$	2.00 m	0.00 m
$\vec{U}$	-5.50 m	2.50 m



7.00 m



5.50 m



2.00 m



13.0 m



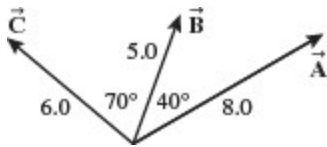
11.1 m



Question 2 5 pts

The figure shows three vectors and their magnitudes and relative directions. The magnitude of the resultant of the three vectors is closest to

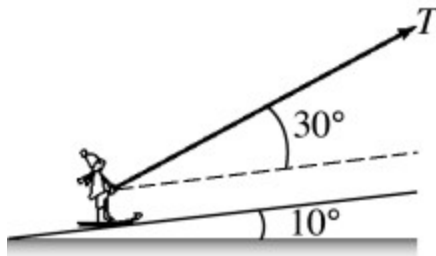
hint: PLace the x-axis along the A vector.



- ☐ 16
- ☐ 13
- ☐ 10
- ☐ 19
- ☐ 7.0
- ☒

Question 3 5 pts

In the figure, a T-bar ski tow pulls a skier up a hill inclined at 10° above horizontal. The skier starts from rest and is pulled by a cable that exerts a tension T at an angle of 30° above the surface of the hill. The mass of the skier is 60 kg and the effective coefficient of kinetic friction between the skis and the snow is 0.100. What is the maximum tension in the cable if the starting acceleration is not to exceed 0.400 g ?



- ☐ 366 N
- ☐ 535 N
- ☐ 246 N
- ☐ 431 N
- ☐ 187 N
- ☒

Question 4 5 pts

A 6.0 kg box slides down an inclined plane that makes an angle of 39° with the horizontal. If the coefficient of kinetic friction is 0.40, at what rate does the box accelerate down the slope?

☐

3.1 m/s^2

☐

3.4 m/s^2

☐

3.7 m/s^2

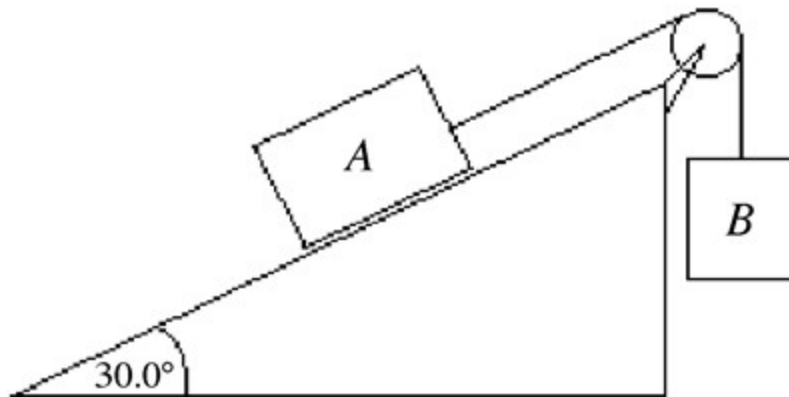
☐

4.1 m/s^2

☐

Question 5 5 pts

Two blocks are connected by a string that goes over an ideal pulley as shown in the figure. Block A has a mass of 3.00 kg and can slide over a rough plane inclined 30.0° to the horizontal. The coefficient of kinetic friction between block A and the plane is 0.400. Block B has a mass of 2.77 kg. What is the acceleration of the blocks? hint: B is going down.


☐

5.35 m/s^2

☐

1.96 m/s^2

☐

0.392 m/s^2

☐

0.00 m/s^2

☐

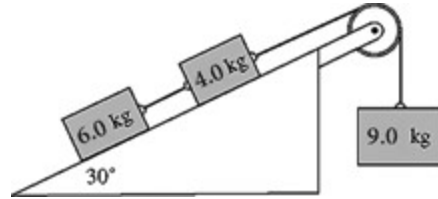
3.12 m/s^2

☐

Question 6 5 pts

A system comprising blocks, a light frictionless pulley, a frictionless incline, and connecting ropes is shown in the figure. The 9.0-kg block accelerates downward when the system is released from rest. The tension in the rope connecting the 6.0-kg block and the 4.0-kg block is closest to.

hint:: first consider 10kg being pulled up hill. Find a. Then consider only the 6kg.



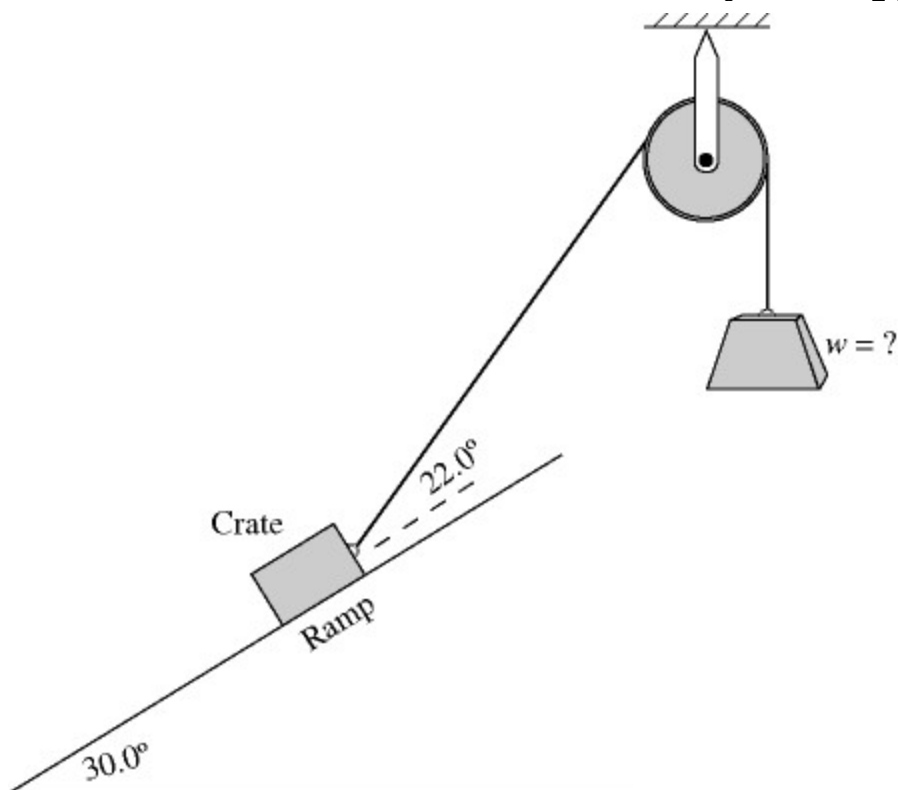
- ☐ 30 N.
- ☐ 33 N.
- ☐ 42 N.
- ☐ 36 N.
- ☐ 39 N.



Question 7 5 pts

A 1520-N crate is to be held in place on a ramp that rises at 30.0° above the horizontal (see figure). The massless rope attached to the crate makes a 22.0° angle above the surface of the ramp. The coefficients of friction between the crate and the surface of the ramp are $\mu_k = 0.450$ and $\mu_s = 0.650$. The pulley has no appreciable mass or friction. What is the MAXIMUM weight w that can be used to hold this crate stationary on the ramp?

hint: you have 3 equations that turns to 2 equations. you can use matrix.



1.



500N



1380 N



2000N

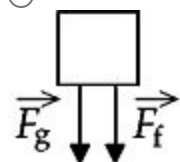


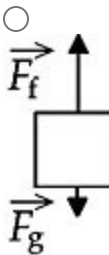
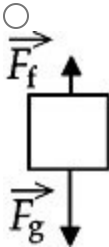
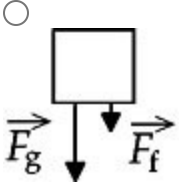
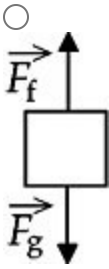
1520N



Question 8 5 pts

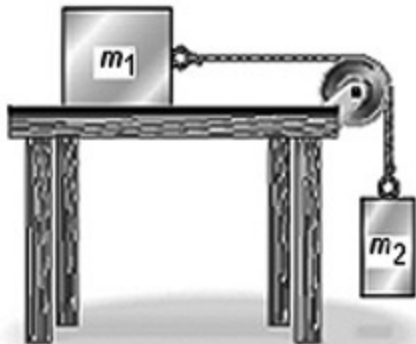
Which one of the following free-body diagrams best represents the free-body diagram, with correct relative force magnitudes, of a person in an elevator that is traveling upward with an unchanging velocity? $\vec{F}_f$ is the force of the floor on the person and $\vec{F}_g$ is the force of gravity on the person.





Question 9 5 pts

Two objects having masses m_1 and m_2 are connected to each other as shown in the figure and are released from rest. There is no friction on the table surface or in the pulley. The masses of the pulley and the string connecting the objects are completely negligible. What must be true about the tension T in the string just after the objects are released?





$$T = m_1 g$$



$$T > m_2 g$$



$$T > m_1 g$$



$$T = m_2 g$$



$$T < m_2 g$$



Question 10 5 pts

A 60.0-kg person rides in an elevator while standing on a scale. The scale reads 400 N. The acceleration of the elevator is closest to



zero.



6.67 m/s² upward.



6.67 m/s² downward.



9.80 m/s² downward.

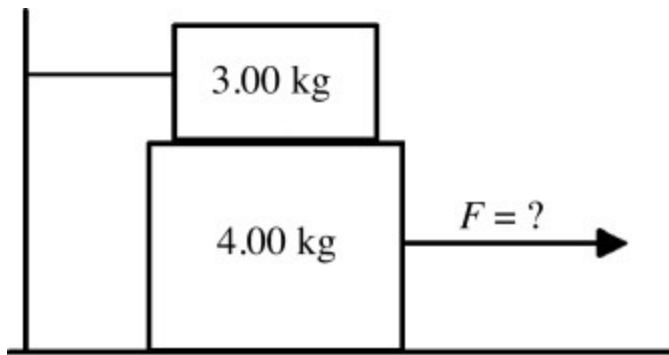


3.13 m/s² downward.



Question 11 5 pts

A 4.00-kg block rests between the floor and a 3.00-kg block as shown in the figure. The 3.00-kg block is tied to a wall by a horizontal rope. If the coefficient of static friction is 0.800 between each pair of surfaces in contact, what horizontal force F must be applied to the 4.00-kg block to make it move? hint: 2 frictional forces are acting on 4kg. From the ground (@ left (depends on 7kg) and from the 3 kg (@ left (depend on 3kg).



- ☐ 23.5 N
- ☐ 78.4 N
- ☐ 54.9 N
- ☐ 21.1 N
- ☐ 16.2 N



Question 12 5 pts

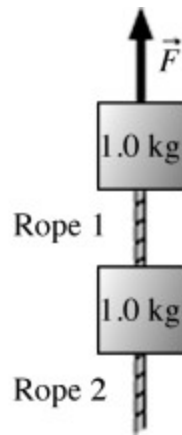
A driver in a 1000 kg car traveling at 20 m/s slams on the brakes and skids to a stop. If the coefficient of friction between the tires and the horizontal road is 0.80, how long will the skid marks be?

- ☐ 21 m
- ☐ 26 m
- ☐ 33 m
- ☐ 24 m



Question 13 5 pts

The figure shows two 1.0 kg-blocks connected by a rope. A second rope hangs beneath the lower block. Both ropes have a mass of 250 g. The entire assembly is accelerated upward at 2.3 m/s^2 by force $\vec{F}$. What is the tension at the top end of rope 1? hint: consider all the masses (including the ropes) being pulled by F and find F . Then consider the top 1 kg mass.


☐

18 N

☐

3.5 N

☐

2.9 N

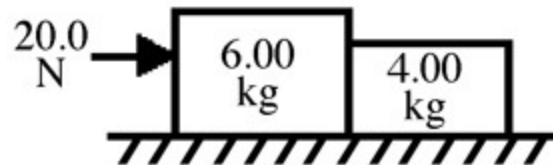
☐

15 N

☒

Question 14 5 pts

A 6.00-kg block is in contact with a 4.00-kg block on a horizontal frictionless surface as shown in the figure. The 6.00-kg block is being pushed by a horizontal 20.0-N force as shown. What is the magnitude of the force that the 6.00-kg block exerts on the 4.00-kg block?


☐

20.0 N

☐

8.00 N

☐

4.00 N

☐

6.00 N

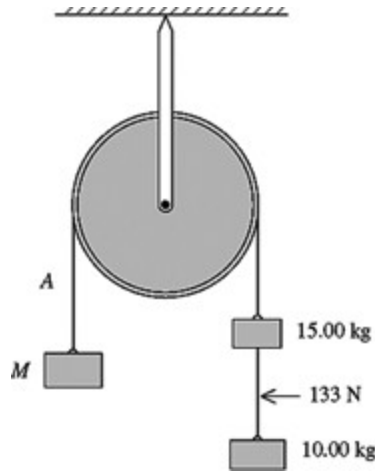
☐

10.0 N



Question 15 5 pts

Three objects are connected by massless wires over a massless frictionless pulley as shown in the figure. The tension in the wire connecting the 10.0-kg and 15.0-kg objects is measured to be 133 N. What is the mass M ? hint: consider the 10kg mass. Since the tension pulling up is larger than the weight ($133 > 100$) that means the system is moving counter clockwise. There is an acceleration. first find the acceleration using the 10kg. ($133\text{N up } 100\text{N down}$). Then consider (10kg+15kg) as one mass to find the tension T in the pulley. Then find M . All have the same acceleration.

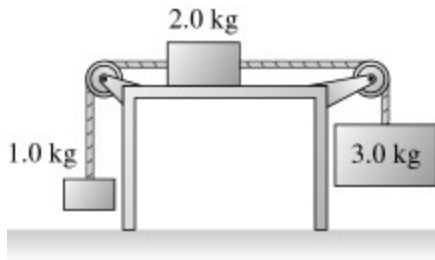


- ☐ 8.33 kg
- ☐ 33.9 kg
- ☐ 95.0 kg
- ☐ 52.8 kg
- ☐ 35.0 kg



Question 16 5 pts

Three objects are connected as shown in the figure. The strings and frictionless pulleys have negligible masses, and the coefficient of kinetic friction between the 2.0-kg block and the table is 0.25. What is the acceleration of the 2.0-kg block?



hint: not hard. obviously it will accelerate down with the 3kg pulling everything and fighting against the friction and the weight of the 1kg. The inertia is the 6kg. If you consider one system. winner - loser = inertia $\times$ a so $3 \times 9.8 - \text{friction} - 1 \times 9.8 = 6a$. friction = μ_k Normal. Normal is the weight of the 2 kg.

☐ 2.5 m/s²

☐ 1.7 m/s²

☐ 3.2 m/s²

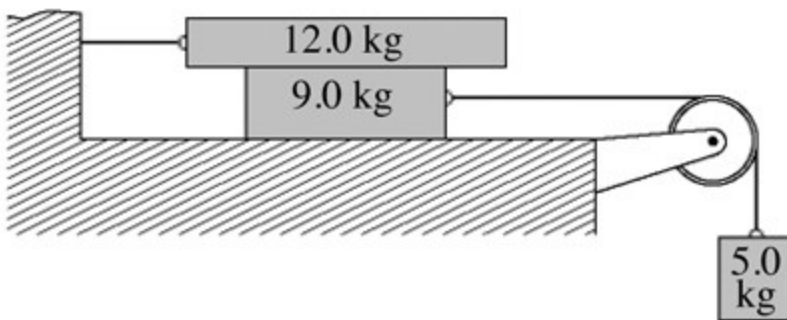
☐ 4.0 m/s²



Question 17 5 pts

A system comprised blocks, a light frictionless pulley, and connecting ropes is shown in the figure. The 9.0-kg block is on a perfectly smooth horizontal table. The surfaces of the 12-kg block are rough, with $\mu_k = 0.30$ between the 2 blocks. If the 5.0-kg block accelerates downward when it is released, find its acceleration.

!! there is no



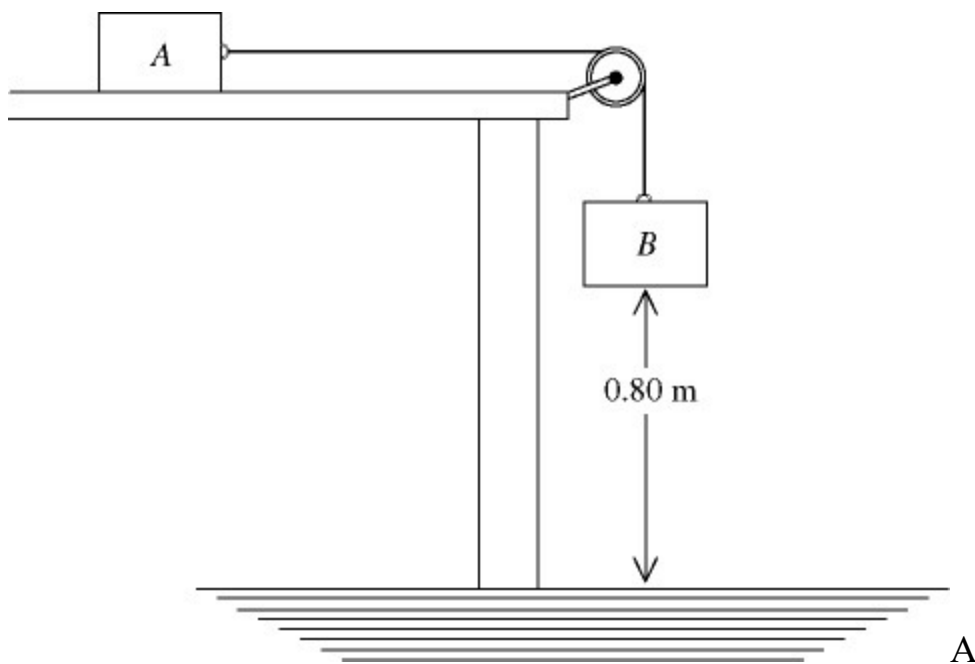
there is a tutorial

☐ 1.0 m/s²

☐1.4 m/s²☐1.8 m/s²☐1.6 m/s²☐3 m/s²☒

Question 18 5 pts

In the figure, two wooden blocks each of 0.30 kg mass are connected by a string that passes over a very light frictionless pulley. One block slides on a horizontal table, while the other hangs suspended by the string, as shown in the figure. At time $t = 0$, the suspended block is 0.80 m over the floor, and the blocks are released from rest. After 2.5 s, the suspended block reaches the floor. What is the coefficient of kinetic friction between the table and the sliding block?



hint: Also easy. First use kinematics to get the acceleration. $y = 0.5 a t^2$

Then B is pulling everything. and you have friction = μ_k Normal on A = $\mu_k m_A g$

So (weight B) - friction = $(m_A + m_B) a$. solve for friction. Then solve for μ_k .

☐

0.35

☐

0.52



0.95



0.84



0.65



Question 19 5 pts

A 6.0 kg box slides down an inclined plane that makes an angle of 39° with the horizontal. If the coefficient of kinetic friction is 0.19, at what rate does the box accelerate down the slope?

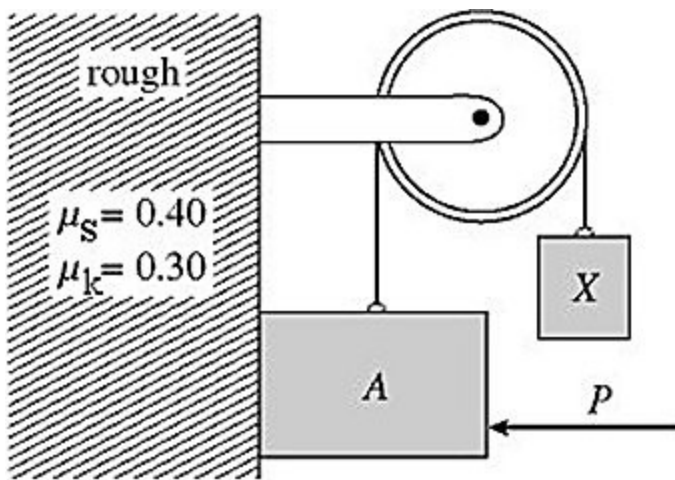
hint: typical inclined plane problem. The x-component of the weight is pulling down slope and friction is resisting. $mg \sin(39) - \text{friction} = m a$. $\text{friction} = \mu_k \text{normal}$.

But $\text{normal} = mg \cos(\text{angle})$ (y-component of the weight).

5.5 m/s^2 5.2 m/s^2 6.2 m/s^2 4.7 m/s^2 

Question 20 5 pts

Block A of mass 5.0 kg and block X are attached to a rope which passes over a pulley, as shown in the figure. An 80-N force P is applied horizontally to block A , keeping it in contact with a rough vertical face. The coefficients of static and kinetic friction between the wall and block A are $\mu_s = 0.40$ and $\mu_k = 0.30$. The pulley is light and frictionless. The mass of block X is adjusted until block A moves upward with an acceleration of 1.6 m/s^2 . What is the mass of block X ?



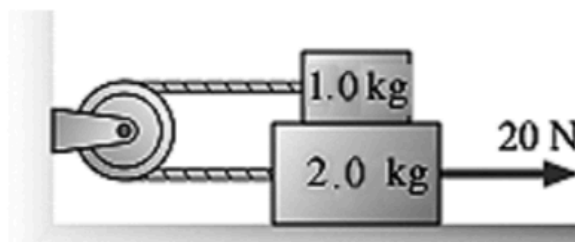
I made a tutorial in youtube/dropbox

- ☐ 8.7 kg
- ☐ 9.9 kg
- ☐ 7.5 kg
- ☐ 8.1 kg
- ☐ 9.3 kg
- ☐

Question 21 4 pts

There is a tutorial in dropbox. Try first

A rope pulls on the lower block in the figure with a tension force of 20 N. The coefficient of kinetic friction between the lower block and the surface is 0.16. The coefficient of kinetic friction between the lower block and the upper block is also 0.16. The pulley has no appreciable mass or friction. What is the acceleration of the 2.0 kg block?



- ☐ 4m/s/s
- ☐ 2m/s/s
- ☐ 1m/s/s



6m/s/s

Not saved

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